Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 0.1 | **Date:** 03-Apr-2026

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1. Administrative & Study Identification

Full Study Title: Phase II Single-Arm Study of Durvalumab and Bevacizumab Following Transarterial Radioembolization Using Yttrium-90 Glass Microspheres (TheraSphere™) in Unresectable Hepatocellular Carcinoma Amenable to Locoregional Therapy **Short Title/Acronym:** A US Study to Evaluate Transarterial Radioembolization (TARE) in Combination With Durvalumab and Bevacizumab Therapy in People With Unresectable Hepatocellular Carcinoma Amenable to TARE (EMERALD-Y90) **Protocol Number:** [Internal Study ID] **NCT Number:** NCT06040099 **Trial Status:** Recruiting **Sponsor Name:** AstraZeneca **Investigational Product:** Transarterial Radioembolization (Yttrium-90 glass microspheres TARE - TheraSphere™), Durvalumab, and Bevacizumab **Phase of Development:** Phase II **Medical Monitor:** [Name and Contact Information]

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To measure the efficacy of transarterial radioembolization (TARE) followed by durvalumab monotherapy and subsequently durvalumab plus bevacizumab, as determined by Progression-Free Survival (PFS) in participants with unresectable HCC amenable to locoregional therapy. **Secondary Objectives:** To evaluate safety and tolerability (Number of participants with adverse events), Objective Response Rate (ORR), and Overall Survival (OS).

2.2 Background & Rationale Intermediate hepatocellular carcinoma (HCC) is often treated with locoregional therapy (LRT), such as TARE. Despite advances in TARE delivery, the median progression-free survival following treatment highlights a pressing need for new treatment options. This study evaluates combining TARE with immune checkpoint inhibitor regimens (durvalumab + bevacizumab) to enhance efficacy and prolong survival in participants with unresectable HCC amenable to embolization.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target N: Approximately 120 participants will be screened to enroll approximately 60 participants. **Number of Sites:** ~20 US sites **Estimated Study Start:** [DD-MMM-YYYY] **Estimated Primary Completion:** [DD-MMM-YYYY]

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Confirmed unresectable Hepatocellular Carcinoma (uHCC) amenable to embolization. 3. Ineligible for or have declined treatment with resection and/or ablation or liver transplant. 4. Child-Pugh score class A and an Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 at enrollment. 5. Lung dose threshold for Yttrium 90 glass microspheres of 30 Gy (≤ 30 Gy per treatment for glass) and an estimated Future liver remnant volume (FLRV) $\ge 30\%$ of whole liver volume. 6. No evidence of extrahepatic disease on any available imaging. 7. One or more measurable lesions; unilobar disease for participants with segmental or right anterior/posterior portal vein invasion (Vp1/Vp2) and eligible for Yttrium 90 glass microspheres TARE. 8. Participants with previous Transarterial Chemoembolization (TACE) or TARE associated with the curative setting are permitted with a 6-month washout. Participants with more than 1 prior embolization are permitted if more than 12 months ago, for a different primary lesion, and FLR > 30%. 9. Adequate organ and marrow function.

4.2 Exclusion Criteria 1. Disease amenable to curative surgery, ablation, or transplantation. Transplant patients are considered eligible if outside of Milan criteria and not currently listed for transplant. 2. Participant has received any prior anticancer systemic therapy for unresectable HCC. 3. Receipt of more than 1 prior embolization (TACE or TARE) treatment/procedure, unless meeting the specific inclusion exception. 4. Clinically significant (eg, active) cardiovascular disease, uncontrolled hypertension, history of hepatic encephalopathy, or a history of an arterial thrombotic event (including myocardial infarction, unstable angina, cerebrovascular accident, or transient ischemic attack) within 6 months prior to enrollment. 5. Participants co-infected with HBV and HDV. 6. Any history of nephrotic or nephritic syndrome. 7. Known hereditary predisposition to bleeding or thrombosis; any prior or current evidence of bleeding diathesis. 8. History of abdominal fistula or gastrointestinal (GI) perforation, non-healed gastric ulcer that is refractory to treatment, or active GI bleeding within 6 months prior to enrollment.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants will receive partition-based dosing of TARE using Yttrium-90 glass microspheres (TheraSphere™). Following TARE, participants will receive durvalumab IV solution followed by maintenance with durvalumab + bevacizumab IV solution (Bevacizumab trade names: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi). **Route of Administration:** Transarterial (for TARE) and Intravenous (IV) solution (for durvalumab and bevacizumab). **Duration of Treatment:**

Until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Medications allowed for standard management of underlying cirrhosis and co-morbidities. **Prohibited:** Other concurrent systemic anti-cancer therapies or investigational agents.

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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Tumor Imaging, Liver Function Tests (Child-Pugh), ECOG PS
Baseline / Day 1	Day 1	Administration of TARE (Y-90 microspheres)
Interim	Cycle [X] $\pm$ [Y]	Administration of Durvalumab (Monotherapy) followed by Durvalumab + Bevacizumab, Safety Labs, Tumor Imaging per mRECIST
End of Study	Follow-up	Final Vital Signs, Exit Interview, Disease Progression Assessment

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS), defined as the time from Day 1 (day of TARE) until the date of progressive disease per modified Response Evaluation Criteria in Solid Tumors (mRECIST), as assessed by the investigator, or death due to any cause. **Measurement Method:** Radiographic Tumor Imaging Assessment per mRECIST. Measured to assess the efficacy of TARE followed by durvalumab monotherapy followed by durvalumab + bevacizumab.

7.2 Secondary Endpoints * Number of participants with Adverse Events (AEs). * Objective Response Rate (ORR). * Overall Survival (OS).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via clinical laboratory evaluations, physical exams, and patient reporting during treatment cycles. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal and/or External safety monitoring committee assessing early safety reviews.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all participants who receive at least one dose of study treatment. **Sample Size Justification:** Target enrollment of approximately 60 participants designed to evaluate the clinical benefit and safety profile of the combination. **Handling of Missing Data:** Standard survival analysis methods (e.g., Kaplan-Meier estimates) with censoring for participants lost to follow-up or without documented progression at the time of data cutoff.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits for source data verification and RECIST image central/local review. **Audit Trail:** Detailed electronic Case Report Form (eCRF) tracking with prompt data clarification handling.

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11. Study Identifiers & Governance

Primary ID: EMERALD-Y90 **Posting ID:** 145568512203123289 **Registry Number:** NCT06040099 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** A US Study to Evaluate Transarterial Radioembolization (TARE) in Combination With Durvalumab and Bevacizumab Therapy in People With Unresectable Hepatocellular Carcinoma Amenable to TARE **Official Regulatory Title:** Phase II Single-Arm Study of Durvalumab and Bevacizumab Following Transarterial Radioembolization Using Yttrium-90 Glass Microspheres (TheraSphere™) in Unresectable Hepatocellular Carcinoma Amenable to Locoregional Therapy

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12. Clinical Framework (Oncology Specific)

Primary Condition: Unresectable Hepatocellular Carcinoma (HCC) **Diagnosis Threshold:** Child-Pugh class A, ECOG 0-1, amenable to embolization **Medical Methodology:** Locoregional Therapy (TARE) + Immune Checkpoint Inhibition + Anti-VEGF **Optimization Target:** Disease progression delay and objective tumor response

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13. Study Procedures & Methodology

Management Pathway: TARE administration followed sequentially by durvalumab monotherapy, then combination immunotherapy **Education Protocol (HCP):** Embolization dosimetry and immunotherapy toxicity management training **Education Protocol (Patient):** Treatment schedule adherence and immune-related adverse event (irAE) symptom recognition **Quality Audit Mechanism:** Periodic early safety reviews and strict mRECIST imaging protocols

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14. Observation & Follow-Up Schedule

Total Duration: Follow-up until disease progression or death **Onsite Visit Cadence:** Aligned with infusion cycles (e.g., Q3W or Q4W for immunotherapy) and imaging schedules **Remote Monitoring Frequency:** Inter-cycle safety checks as required **Checklist Review Interval:** Every imaging cycle

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]